

Climate and Health: Teaching Synthesis

Evidence-based teaching resource

Evidence checked 2026-08-09

1 Currentness Notice

Sources checked on 2026-08-09. Dataset release labels, refresh dates, and component observation periods are not interchangeable; surveillance indicators may not represent complete population incidence.

2 Purpose and Scope

This chapter presents a concise, source-attributed teaching synthesis of Climate and Health. It distinguishes regulations, proposals, guidance, scientific assessments, and datasets.

3 Evidence and Authority Framework

A defensible application identifies the issuing organization, jurisdiction, effective status, relevant date, population or geographic scope, and stated limitations before using a claim.

4 Current Authoritative Baseline

Centers for Disease Control and Prevention CDC's BRACE framework uses five iterative steps: anticipate impacts and assess vulnerabilities; project disease burden; assess interventions; develop and implement an adaptation plan; and evaluate impacts and improve activities.

Centers for Disease Control and Prevention CDC's climate-health resource page links the Heat & Health Tracker and Tracking Network indicators rather than serving as the underlying dataset.

Centers for Disease Control and Prevention CDC Tracking documents historical and projected heat, heat-related emergency visits, hospitalizations, mortality, and vulnerability/preparedness indicators from multiple sources.

Centers for Disease Control and Prevention Heat-related illness surveillance may miss cases when excess heat is not explicitly documented and should not be interpreted as complete population incidence.

Centers for Disease Control and Prevention The Heat & Health Tracker uses NSSP syndromic surveillance; its primary health source is heat-related emergency-department visits reported as rates per 100,000 ED visits.

Agency for Toxic Substances and Disease Registry The HHI is a national ZIP Code Tabulation Area-level index combining historical heat-health burden, sensitivity, sociodemographic, and natural/built-environment modules.

Agency for Toxic Substances and Disease Registry HHI uses ZCTAs to approximate ZIP-code geography and ranks indicators within four modules plus an overall rank.

Agency for Toxic Substances and Disease Registry The 2024 HHI covers the 48 contiguous states and District of Columbia; Alaska, Hawaii, Puerto Rico, and other U.S. territories are excluded because required indicator coverage was insufficient.

Agency for Toxic Substances and Disease Registry The current registered HHI data release is labeled 2024; that release label is distinct from the various observation periods of its component indicators.

World Health Organization WHO identifies age, health status, occupation, socioeconomic conditions, acclimatization, infrastructure, and local conditions as factors shaping heat vulnerability and impact.

World Health Organization Regional Office for Europe WHO/Europe's 2026 second-edition heat-health action-plan guidance is organized around eight core elements spanning governance, warning systems, vulnerable populations, communication, health-system resilience, exposure reduction, surveillance, and evaluation.

World Health Organization WHO describes climate-health pathways through extreme weather, heat, air and water quality, food systems, infectious disease, mental health, displacement, and health-system disruption.

Centers for Disease Control and Prevention CDC/ATSDR's strategic framework emphasizes health equity, surveillance and analytics, research and evaluation, preparedness/adaptation, and education/dissemination.

U.S. Global Change Research Program NCA5 Chapter 15 assesses interacting climate-health pathways including heat, air quality, infectious disease, water and food risks, mental health, and health-system effects.

U.S. Global Change Research Program NCA5 emphasizes that exposure, sensitivity, and adaptive capacity are unevenly distributed and that climate change can worsen health inequities.

U.S. Global Change Research Program NCA5 Chapter 30 provides region-specific evidence for Hawaii and U.S.-Affiliated Pacific Islands; findings should not be generalized uniformly across islands or communities.

5 Instructional Application

Learners classify source authority, test the boundaries of each claim, construct a hypothetical professional decision, and communicate what is known, proposed, recommended, or unresolved.

6 Common Errors

- Treating proposed regulatory text as current final law.
- Treating recommendations or guidance as automatically enforceable.
- Generalizing regional or dataset-specific findings beyond their documented scope.
- Confusing publication, release, effective, retrieval, and observation-period dates.

7 Limitations

This synthesis is not a legal opinion or an operational substitute for current agency text, site-specific professional judgment, or applicable state and local requirements.

8 Conclusion

Professional teaching is strengthened when factual claims are current, authority distinctions remain visible, uncertainty is explicit, and external sources are independently identifiable.

9 References

- U.S. Global Change Research Program. Fifth National Climate Assessment, Chapter 15: Human Health. <https://nca2023.globalchange.gov/chapter/15/>
- U.S. Global Change Research Program. Fifth National Climate Assessment, Chapter 30: Hawaii and U.S.-Affiliated Pacific Islands. <https://nca2023.globalchange.gov/chapter/30/>
- Centers for Disease Control and Prevention. Building Resilience Against Climate Effects framework. <https://www.cdc.gov/climate-health/php/brace/index.html>
- Centers for Disease Control and Prevention. Climate and Health Program data and tools. <https://www.cdc.gov/climate-health/php/resources/data-and-tools.html>
- Centers for Disease Control and Prevention. Environmental Public Health Tracking heat and climate data documentation. <https://www.cdc.gov/environmental-health-tracking/php/data-research/heat-heat-related-illness.html>
- Centers for Disease Control and Prevention. Heat and Health Tracker source documentation. <https://www.cdc.gov/nssp/php/partnerships/cdc-heat-health-tracker-uses-nssp-data.html>

- Agency for Toxic Substances and Disease Registry. Heat and Health Index landing page. <https://www.atsdr.cdc.gov/place-health/php/hhi/index.html>
- Agency for Toxic Substances and Disease Registry. Heat and Health Index technical documentation. <https://stacks.cdc.gov/view/cdc/158548>
- Agency for Toxic Substances and Disease Registry. Latest Heat and Health Index downloadable dataset and data dictionary. <https://www.atsdr.cdc.gov/place-health/php/hhi/index.html>
- World Health Organization. Heat and health fact sheet. <https://www.who.int/news-room/fact-sheets/detail/climate-change-heat-and-health>
- World Health Organization Regional Office for Europe. Heat-health action plans: guidance, second edition. <https://www.who.int/europe/publications/i/item/9789289062930>
- World Health Organization. Climate change and health fact sheet. <https://www.who.int/news-room/fact-sheets/detail/climate-change-and-health>
- Centers for Disease Control and Prevention. Climate and Health Strategic Framework. <https://www.cdc.gov/climate-health/php/about/strategic-framework.html>